Functional hierarchy for independent electrical review. PCB routing, DIM-003, stackup/copper, DRC and Review B remain open.

Input protection and monitor

3V3_DIGITAL
GND_PWR
I2C3_SDA
VBAT_SYS
FAULT
I2C3_SCL

Файл: PCB-PWR_01_INPUT_PROTECTION.kicad_sch

3V8 modem rail

3V3_DIGITAL
EN_MODEM
VBAT_SYS
3V8_MODEM
GND_PWR

Файл: PCB-PWR_02_3V8_MODEM.kicad_sch

3V3 digital rail

3V3_DIGITAL
GND_PWR
I2C3_SDA
VBAT_SYS
FAULT
I2C3_SCL
PWR_GOOD

Файл: PCB-PWR_03_3V3_DIGITAL.kicad_sch

1V8 auxiliary and MAIN harness

3V3_DIGITAL
EN_MODEM
GND_PWR
I2C3_SDA
3V8_MODEM
FAULT
I2C3_SCL
PWR_GOOD

Файл: PCB-PWR_04_AUX_HARNESS.kicad_sch

NOT FOR MANUFACTURE – routing/DRC/DFM/Review B pending
Root sheet 1 of 5
Human-readable hierarchy; exact pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /
File: PCB-PWR.kicad_sch

Title: PCB-PWR system overview

Size: A3	Date: 2026-09-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/5

Explicit wire stubs expose every reviewed pin/net connection. Hierarchical labels are the only cross-sheet connections.

J1
VBAT_RAW 1 VBAT_PLUS
GND_PWR 2 VBAT_MINUS
PWR_INPUT_43045-0213

F1
VBAT_RAW 1 2 VBAT_FUSED
0451008.MRL

U1
LM74700QDBVRQ1
VBAT_FUSED 6 ANODE
VBAT_FUSED 3 EN
LM74700_VCAP 1 VCAP
GND 2
GND_PWR
CATHODE 4 VBAT_PROTECTED
GATE 5 REV_GATE

Q1
CSD18540Q5B
VBAT_FUSED 1 SOURCE
VBAT_FUSED 2 SOURCE
VBAT_FUSED 3 SOURCE
GATE 4
REV_GATE
DRAIN 5 VBAT_PROTECTED
DRAIN 6 VBAT_PROTECTED
DRAIN 7 VBAT_PROTECTED
DRAIN 8 VBAT_PROTECTED

D1
VBAT_FUSED 1 2 GND_PWR
SMBJ18A CANDIDATE UNIDIRECTIONAL TVS

C1
LM74700_VCAP 1 2 VBAT_FUSED
100 nF 25 V X7R CGA2B3X7R1E104K050BB

C9
VBAT_FUSED 1 2 GND_PWR
100 nF 50 V X7R CGA3E2X7R1H104K080AA

C10
VBAT_PROTECTED 1 2 GND_PWR
4.7 uF 50 V X7R CGA6P3X7R1H475K250AB

RSH1
VBAT_PROTECTED 1 CURRENT_SOURCE
SHUNT_SOURCE_SENSE 3 SENSE_SOURCE
CURRENT_LOAD 2 VBAT_SYS
SENSE_LOAD 4 SHUNT_LOAD_SENSE
WSK2512R0100FEA 10mOhm 1% 1W 4T

U2
INA226AIDGSR
3V3_DIGITAL
SHUNT_SOURCE_SENSE 10 IN+
VBAT_SYS 8 VBUS
I2C2_SCL 5 SCL
I2C2_SDA 4 SDA
VS 6
IN- 9 SHUNT_LOAD_SENSE
ALERT 3 FAULT
A1 1 GND_PWR
A0 2 GND_PWR
GND 7 GND_PWR

C2
3V3_DIGITAL 1 2 GND_PWR
100 nF 25 V X7R CGA2B3X7R1E104K050BB

C13
VBAT_SYS 1 2 GND_PWR
100 uF 35 V hybrid EEH-ZK1V101XP

TP1
VBAT_PROTECTED 1 TEST
TP_VBAT_PROTECTED DFT_PAD_1P70_NOPASTE_REV_A

TP2
SHUNT_SOURCE_SENSE 1 TEST
TP_SHUNT_SRC DFT_PAD_1P70_NOPASTE_REV_A

TP3
SHUNT_LOAD_SENSE 1 TEST
TP_SHUNT_LOAD DFT_PAD_1P70_NOPASTE_REV_A

TP8
FAULT 1 TEST
TP_FAULT DFT_PAD_1P70_NOPASTE_REV_A

TP9
I2C2_SCL 1 TEST
TP_SCL DFT_PAD_1P70_NOPASTE_REV_A

TP10
I2C2_SDA 1 TEST
TP_SDA DFT_PAD_1P70_NOPASTE_REV_A

#FLG01
VBAT_SYS 1 PWR_SOURCE
ERC_SOURCE_FLAG

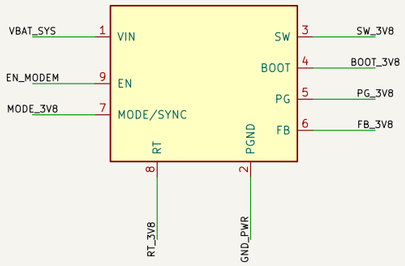
#FLG02
GND_PWR 1 PWR_SOURCE
ERC_SOURCE_FLAG

NOT FOR MANUFACTURE — routing/DRC/DFM/Review B pending
Functional sheet 2 of 5
Human-readable hierarchy; exact pin/net equivalence enforced
Dioneya / ZS-BPLA
Sheet: /Input protection and monitor/
File: PCB-PWR_01_INPUT_PROTECTION.kicad_sch
Title: Input protection + monitor
Size: A3 | Date: 2026-09-17 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 2/5

Explicit wire stubs expose every reviewed pin/net connection. Hierarchical labels are the only cross-sheet connections.

U3

LMR604403SRAKR 3V8



L1

SW_3V8 1 2 3V8_MODEM

XAL7030-472MEC 4.7uH

C11

VBAT_SYS 1 2 GND_PWR

4.7 uF 50 V X7R CGA6P3X7R1H475K250AB

C4

BOOT_3V8 1 2 SW_3V8

100 nF 25 V X7R CGA2B3X7R1E104K050BB

R1

3V8_MODEM 1 2 FB_3V8

100 kOhm 0.1% ERA-2AEB104X

R2

FB_3V8 1 2 GND_PWR

35.7 kOhm 0.1% ERA-2AEB3572X

C3

3V8_MODEM 1 2 GND_PWR

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C14

3V8_MODEM 1 2 GND_PWR

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C15

3V8_MODEM 1 2 GND_PWR

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C16

3V8_MODEM 1 2 GND_PWR

22 uF 25 V X7R CGA6P3X7R1E226M250AB

R3

RT_3V8 1 2 GND_PWR

86.6 kOhm 0.1% ERA-2AEB8662X

R4

MODE_3V8 1 2 GND_PWR

0 ohm ERJ-2GE0R00X

R5

MODE_3V8 1 2 RT_3V8

0 ohm ERJ-2GE0R00X

R6

EN_MODEM 1 2 GND_PWR

100 kOhm 1% ERJ-2RKF1003X

R15

PG_3V8 1 2 3V3_DIGITAL

10 kOhm 1% ERJ-2RKF1002X

TP4

3V8_MODEM 1 TEST

TP_3V8 DFT_PAD_1P70_NOPASTE_REV_A

TP5

PG_3V8 1 TEST

TP_PG_3V8 DFT_PAD_1P70_NOPASTE_REV_A

C20

VBAT_SYS 1 2 GND_PWR

100 nF 50 V X7R CGA3E2X7R1H104K080AA

NOT FOR MANUFACTURE – routing/DRC/DFM/Review B pending
Functional sheet 3 of 5
Human-readable hierarchy; exact pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /3V8 modem rail/
File: PCB-PWR_02_3V8_MODEM.kicad_sch

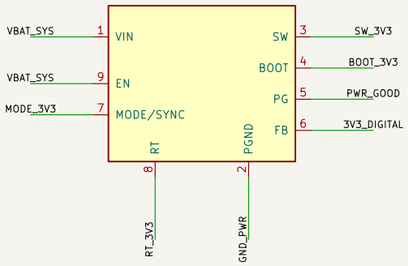
Title: 3V8 modem rail

Size: A3	Date: 2026-09-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/5

Explicit wire stubs expose every reviewed pin/net connection. Hierarchical labels are the only cross-sheet connections.

U4

LMR604403SRAKR 3V3 AON



L2

XAL7030-472MEC 4.7uH

C12

4.7 uF 50 V X7R CGA6P3X7R1H475K250AB

C6

100 nF 25 V X7R CGA2B3X7R1E104K050BB

R10

10 kOhm 1% ERJ-2RKF1002X

C5

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C17

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C18

22 uF 25 V X7R CGA6P3X7R1E226M250AB

C19

22 uF 25 V X7R CGA6P3X7R1E226M250AB

R7

86.6 kOhm 0.1% ERA-2AEB8662X

R8

0 ohm ERJ-2GE0R00X

R9

0 ohm ERJ-2GE0R00X

R12

10 kOhm 1% ERJ-2RKF1002X

R13

4.7 kOhm 1% ERJ-2RKF4701X

R14

4.7 kOhm 1% ERJ-2RKF4701X

TP6

TP_PWR_GOOD DFT_PAD_1P70_NOPASTE_REV_A

#FLG03

ERC_SOURCE_FLAG

C21

100 nF 50 V X7R CGA3E2X7R1H104K080AA

NOT FOR MANUFACTURE – routing/DRC/DFM/Review B pending
Functional sheet 4 of 5
Human-readable hierarchy; exact pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /3V3 digital rail/
File: PCB-PWR_03_3V3_DIGITAL.kicad_sch

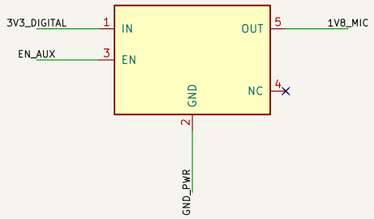
Title: 3V3 AON rail

Size: A3	Date: 2026-09-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/5

Explicit wire stubs expose every reviewed pin/net connection. Hierarchical labels are the only cross-sheet connections.

U5

TPS7A2018PDBVR 1V8_MIC



C7



2.2 uF 10 V X7R CGA3E1X7R1A225K080AC

C8



2.2 uF 10 V X7R CGA3E1X7R1A225K080AC

R11

100 kOhm 1% ERJ-2RKF1003X

NT1

2-pad copper net tie NET_TIE_COPPER_REV_A

NT2

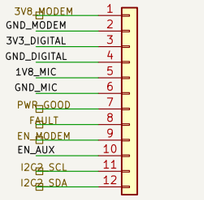
2-pad copper net tie NET_TIE_COPPER_REV_A

NT3



2-pad copper net tie NET_TIE_COPPER_REV_A

J2



MAIN_PWR_43045-1202_12PIN

TP7



TP_1V8 DFT_PAD_1P70_NOPASTE_REV_A

NOT FOR MANUFACTURE – routing/DRC/DFM/Review B pending
Functional sheet 5 of 5
Human-readable hierarchy; exact pin/net equivalence enforced
Dioneya / ZS-BPLA

Sheet: /1V8 auxiliary and MAIN harness/
File: PCB-PWR_04_AUX_HARNESS.kicad_sch

Title: 1V8 + MAIN harness

Size: A3	Date: 2026-09-17	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/5